

Reference excerpt 01-038

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The delay ta_{QP} is measured over the JVP+ECG trace as shown in Fig. 2. The distance IG is measured on the volunteers chest. The velocity c is the ratio between IG and ta_{QP} . Compliance calculation From the Moens-Korteweg equation we obtain the compliance per unit length: $C = CSA \times c^2$ (5) where CSA_x is the CSA of the IJV measured in G at the x wave. Pressure gradient calculation The pressure inside the IJV is calculated from the instantaneous CSA as follows: $p(t) = \frac{1}{C} CSA(t)$ (6) substituting the expression of $p(t)$ obtained above into Eq. (2) we obtain the expression for the pressure gradient: $p_z = \frac{1}{C} \frac{dCSA}{dt}$ (7) Flow velocity calculation $w(t, z)$ The function $w(t, z)$ is calculated by inserting into the Eq.(1) the expression for the pressure gradient found in Eq. (7). The mathematical details are given in [Sisini et al.(2015b)]. 3 Figure 3: Calculated velocity trace together with the experimental one obtained by the Doppler examination. RESULTS EXAMPLE Pressure waves velocity calculation The measured delay (ta_{QP}) was 0.14 s and the distance IG was 23 cm. As a consequence the velocity c was 164 cm/s. Compliance calculation The minimum value of the CSA (CSA_x) during the cardiac cycle was 0.2 cm² and the compliance for unit of length was 9.8 10⁻³ cm²/mmHg. Flow velocity calculation $w(t, z)$ The velocity was calculated following Eq. (1) and its trace is shown in Fig. 3 together with the experimental trace